
Predicting Actual Body Weight Based on Fitness, Physiological and Demographic Data

Melvin Lopez

NetID: mnl90

Section: 05

Brian Vega-Ortega

NetID: bv226

Section: 05

Abstract

Predicting body weight and understanding factors associated with physical fitness are important topics in health and data science. In this project, we study how accurately a person's actual body weight can be predicted using exercise behavior, physiological measurements, and lifestyle-related indicators from a noisy real-world-inspired fitness dataset. The dataset includes calories burned, dream weight, age, gender, workout duration, heart rate, body mass index (BMI), weather conditions, and exercise intensity. To simulate real-world conditions, the dataset contains missing values, inconsistent labels, duplicate entries, and outliers.

We preprocess the data by handling missing values, correcting inconsistencies, encoding categorical variables, scaling numerical features, and separating the target variable. Exploratory data analysis is used to identify relationships between variables and body weight. We train multiple regression models and evaluate them using MAE, MSE, RMSE, and R^2 .

Results show that exercise patterns, BMI, heart rate, and related features contribute to predicting body weight, while also highlighting the importance of robust pre-processing on noisy datasets.

Code: <https://github.com/MelvinBrianFinalProject/Predicting-Actual-Body-Weight-Based-on-Fitness-Framework/tree/main>

Video: <https://drive.google.com/file/d/1Enm3tRq9a2BuHZYTn3P1QJASop8WvfQb/view?usp=sharing>

1 Introduction

Body weight and physical fitness are important indicators of overall health and wellness. Many people track exercise habits, calories burned, heart rate, and body composition to understand physical progress. With increasing availability of fitness tracking data, machine learning can be applied to identify relationships between physiological measurements, exercise behavior, and body weight outcomes.

The main research question is how accurately body weight can be predicted using exercise behavior, physiological metrics, and fitness-related indicators. This is important because fitness tracking systems are widely used to monitor health progress, and understanding key predictive factors may provide insight into physical outcomes.

In this project, we use a fitness dataset containing exercise, physiological, and demographic variables. We perform preprocessing and exploratory data analysis, then train multiple regression models and compare their predictive performance using standard evaluation metrics. Prior work by Fujihara et al. (2023) shows that machine learning can

estimate body weight using health-related data. The goal is not only prediction but also identifying which variables are most strongly associated with body weight.

2 Related Work

Exercise behaviors, physiological measurements, and body composition have been widely studied in health science, fitness analytics, and sports science. Prior research generally shows that factors such as workout duration, exercise intensity, calories burned, heart rate, and body composition all contribute to the overall physical health and fitness outcomes. Body weight is influenced by many factors, including exercise habits, calories, body composition, and nutrition. In a large meta-analysis on resistance training and protein supplementation, Morton et al. reported that protein supplementation “did not have a significant effect on changes in TBM,” meaning total body mass (Morton et al., 2018). This supports the idea that body weight is not explained by one factor alone, but by a combination of training behavior, nutrition, and individual physical characteristics. Body weight and physical fitness are influenced by multiple interconnected factors rather than a single variable. These include age, gender, activity level, workout consistency, recovery, and physiological characteristics.

2.1 Exercise behavior and body composition

Exercise behavior is strongly connected to changes in body composition and overall fitness. Individuals who consistently participate in physical activity often experience differences in calories burned, cardiovascular performance, and body weight outcomes compared to sedentary individuals. Workout duration and exercise intensity may also influence physical progress because higher-intensity exercise generally requires greater energy to be expended.

In many fitness studies, body composition indicators such as BMI are commonly used to estimate relationships between exercise habits and physical health outcomes. However, these measurements are affected by multiple variables and may not fully explain individual fitness progress on their own.

2.2 Physiological measurements and fitness indicators

Physiological indicators such as heart rate, BMI, and calorie expenditure are commonly used to monitor performance and overall health. Heart rate can reflect exercise intensity and cardiovascular response during physical activity, while calories burned may indicate overall energy expenditure. These measurements are often analyzed together to better understand exercise behavior and fitness patterns.

Environmental and demographic factors may also influence physical outcomes. Variables such as age, gender, and weather conditions can affect workout performance, recovery, and exercise efficiency. Because of this, fitness prediction problems often require analyzing multiple interacting variables rather than examining individual measurements independently.

2.3 Machine learning in fitness analysis

Machine learning techniques are increasingly used in fitness and health analytics to identify patterns in physiological and exercise-related data. Regression models can predict continuous outcomes such as body weight or calories expenditure, while classification models can group individuals based on fitness level or activity patterns.

These methods are useful because they can capture relationships between variables that may not be obvious through simple observation alone. More advanced machine learning models can also identify nonlinear relationships and estimate feature importance, helping researchers determine which variables contribute most strongly to predictive performance

2.4 Our contribution

This project applies a data-driven machine learning approach to predict actual body weight using exercise behavior, physiological measurements, and fitness-related indicators. Unlike studies that focus on a single fitness factor, this project examines how multiple variables such as calories burned, BMI, workout duration, heart rate, exercise intensity, and demographic characteristics interact in predicting body weight outcomes.

Additionally, this project emphasizes robust pre-processing techniques by working with a noisy version of the dataset containing missing values, inconsistent formatting, duplicate entries, and outliers. Multiple regression models are evaluated and compared to analyze predictive performance and feature importance across the fitness-related variables.

3 Methodology

3.1 Dataset description

The dataset used in this project is a noisy, real-world-inspired fitness dataset containing 3,883 cleaned observations after pre-processing. It includes information about individuals' exercise behavior, physiological measurements, and demographic attributes. The dataset includes variables such as calories burned, dream weight, age, gender, workout duration, heart rate, body mass index (BMI), weather condition, and exercise intensity.

Each record includes exercise, physiological, and demographic variables such as calories burned, dream weight, age, gender, workout duration, heart rate, BMI, exercise intensity, and weather conditions. The dataset contains real-world imperfections including missing values, inconsistent labels, duplicate entries, and outliers.

The target variable for this project is **Actual Body Weight**. This variable represents the continuous value describing a participant's body weight that the machine learning models attempt to predict based on the available exercise, physiological, and lifestyle-related features.

3.2 Data pre-processing

Before applying machine learning models, the dataset was cleaned and prepared to ensure reliability and consistency. Missing values were identified and handled using appropriate strategies depending on the feature type. Rows with missing critical values were removed or imputed when able. Categorical inconsistencies were corrected by standardizing labels and duplicate rows were removed to prevent bias in training. Categorical variables including Exercise type, Gender, and Weather Conditions were encoded using appropriate encoding methods such as one-hot encoding or label encoding.

Numerical variables including Calories Burned, Dream Weight, Age, Duration, Heart Rate, BMI and Exercise Intensity were used directly after cleaning, and scaling was applied where necessary to ensure consistent feature magnitude across models.

Finally, the dataset was split into training and testing sets: Training set: 3,106 samples Testing set: 777 samples

3.3 Exploratory data analysis

Exploratory data analysis (EDA) was performed to better understand the distribution and relationships between variables. The distribution of **Actual Body Weight** was analyzed to understand the range and variability of body weight across individuals. The dataset shows weights ranging from approximately 45.78 to 216.60, with a mean of 75.64 and a standard deviation of 16.02, indicating moderate variability with some high-end outliers.

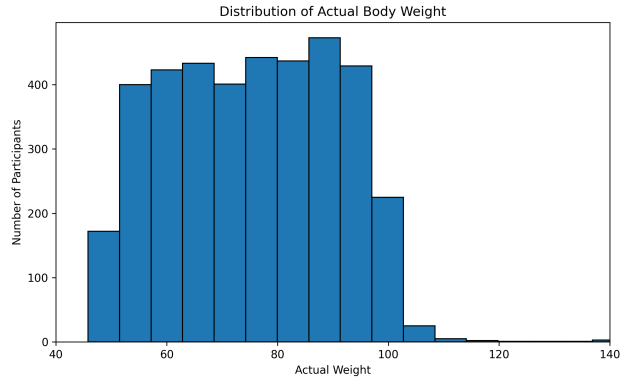


Figure 1: Distribution of actual body weight in the dataset. This figure shows the spread of body weight values across participants

Figure 1 illustrates that most participants fall within moderate body weight range, while a smaller number of observations appear at higher weight values. Identifying these outliers is important because extreme values may influence regression model performance

Additionally, relationships between key features such as BMI, dream weight, calories burned, heart rate, and workout duration were analyzed to determine their association with body weight.

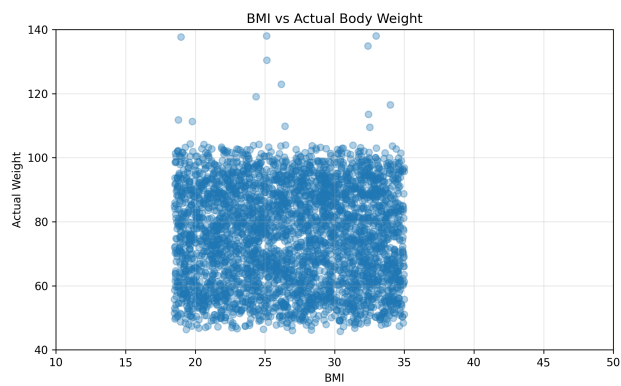


Figure 2: Relationship between BMI and actual body weight.

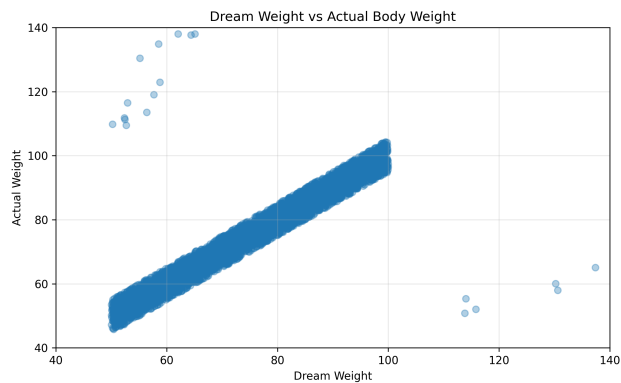


Figure 3: Relationship between dream weight and actual body weight.

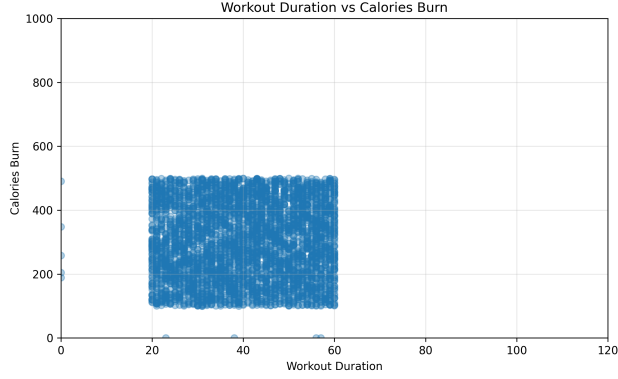


Figure 4: Relationship between workout duration and calories burned.

3.4 Model selection

Since the target variable Actual Weight is continuous, regression-based machine learning models were used. Multiple models were trained and evaluated to compare predictive performance and determine which model best captured the relationships between exercise behavior, physiological measurements, and body weight.

3.4.1 Linear regression

Linear Regression was used as a baseline model because it is simple, computationally efficient, and easy to interpret. The model assumes a linear relationship between the input features and the target variable.

The general form of the model is:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p \quad (1)$$

where $\hat{y}$ is the predicted actual body weight, $x_1, x_2, \dots, x_p$ are the input features, and $\beta_0, \beta_1, \dots, \beta_p$ are the learned model coefficients.

Linear Regression provides a useful baseline for comparing more advanced machine learning models

3.4.2 Decision tree Regressor

A Decision Tree model was also evaluated because it can capture nonlinear relationships between variables such as BMI, calories burned, heart rate, and workout duration. Unlike Linear Regression, Decision Trees do not assume that relationships between variables are strictly linear.

Instead, the model recursively splits the dataset into smaller regions based on feature thresholds and predicts values using learned decision rules. This allows the model to better capture complex interactions between exercise-related and physiological variables.

3.4.3 Random Forest Regressor

Random Forest Regression was used as an ensemble learning model. It builds multiple decision trees using random subsets of the training data and combines their predictions to improve overall performance and reduce overfitting.

This model is especially useful for fitness-related prediction tasks because it can model nonlinear interactions between variables while maintaining strong generalization performance. Random Forest also provides feature importance scores, which help identify which variables contribute most strongly to predicting actual body weight.

4 Evaluation

4.1 Evaluation metrics

The regression models were evaluated using standard regression metrics. These metrics help measure how closely the predicted values match the actual body weight values.

- Mean Absolute Error (MAE), Measures the average absolute difference between predicted and actual values.
- Mean Squared Error (MSE), Measures the average squared prediction error and penalizes larger errors more heavily.
- Root Mean Squared Error (RMSE), Represents the prediction error in the same unit as the target variable.
- R^2 Score, Measures the proportion of variance in the target variable is explained by the model.

4.2 Model results

Table 1 shows the performance comparison of the machine learning models on the testing dataset.

Table 1: Performance comparison of regression models on the test dataset.

Model	MAE	MSE	RMSE	R^2
Linear Regression	4.583299	94.213964	9.706388	0.648558
Decision Tree Regressor	3.636091	91.482540	9.564651	0.658747
Random Forest Regressor	3.479053	74.615756	8.638041	0.721665

Overall, the best-performing model was **Random Forest Regressor**. It produced the lowest prediction errors and the highest R^2 score, indicating that it was the most effective model for predicting the actual body weight in this dataset

4.3 Prediction visualization

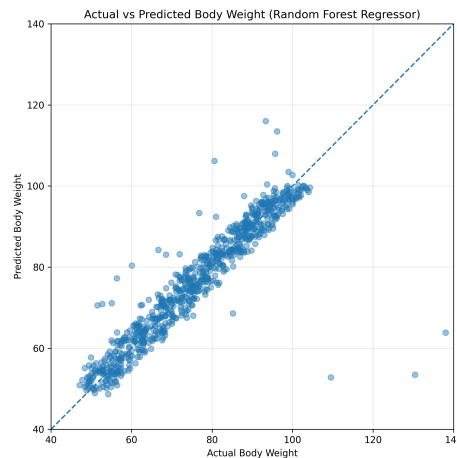


Figure 5: Actual versus predicted body weight for the best-performing model (Random Forest Regressor). Points closer to the diagonal line indicate more accurate predictions.

Figure 5 compares the actual values with the predicted values from the best-performing model.

4.4 Feature importance

Table 2: Top feature importance rankings from the Random Forest model.

Feature	Importance
Dream Weight	0.8846
BMI	0.0216
Calories Burned	0.0155
Age	0.0144
Exercise Intensity	0.0114
Heart Rate	0.0106
Workout Duration	0.0100

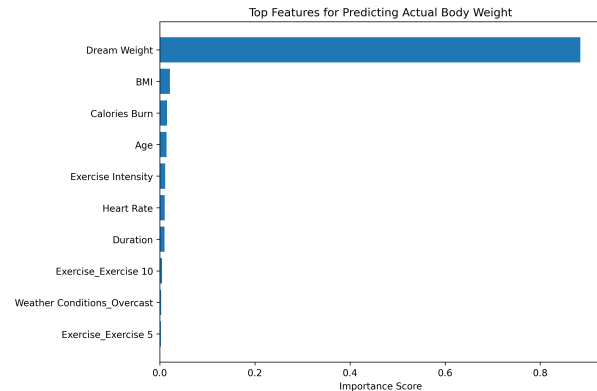


Figure 6: Feature importance from the Random Forest Regressor showing contribution of each variable in predicting actual body weight.

Table 2 shows which features contributed most strongly to the Random Forest model’s predictions. Among all features, Dream Weight had by far the largest importance score, indicating that it was the strongest predictor of actual body weight in the dataset. Variables such as BMI, calories burned, exercise intensity, heart rate, and workout duration, also contributed to the model although it was to a smaller degree.

5 Discussion

The results suggest that body weight can be predicted using physiological measurements, exercise behavior, and demographic information. Variables such as BMI, calories burned, heart rate, workout duration, and exercise intensity contribute to predictive performance.

Among all features, Dream Weight had the strongest influence on the Random Forest model. This is expected since it reflects a target or self-reported weight. However, physiological and exercise-related variables still improve predictive performance.

The Random Forest Regressor model performed the best out of both Linear Regression and Decision Tree Regression. This provides insight in a way that the relationships between the features and actual body weight are not solely linear. Fitness-related variables interact in complex ways, and models such as Random Forest capture these nonlinear relationships better.

Additionally, the project highlights the importance of robust pre-processing when working with noisy datasets. Handling missing values, inconsistent labels, duplicate entries, and outliers was important in improving model reliability and predictive performance.

6 Limitations

This project has several limitations. First, the dataset may not fully represent all populations since body weight and fitness outcomes are influenced by many personal and environmental factors. Second, important variables such as quality of sleep, diet quality, stress levels, and long-term exercise consistency were not included in the dataset.

Another limitation is that some of the data may be simulated or self-reported which can introduce errors or bias in the recordings. In addition, correlation between variables does not imply causation. Even if variables such as BMI,

calories burned, or dream weight are strongly associated with actual body weight, this does not prove that they directly cause changes in body weight.

A further limitation is that dream weight was overwhelmingly the most important factor in the Random Forest model. This did improve predictive accuracy however it may have also reduced the relative contribution of other exercise-related variables, potentially dominating the prediction process.

Finally, although the Random Forest model achieved the strongest overall performance, there is still unexplained variation in the target variable suggesting that additional unmeasured factors may influence body weight prediction.

7 Conclusion

This project explored the use of machine learning techniques to predict actual body weight using exercise behavior, physiological measurements, and demographic information from a noisy dataset.

After pre-processing and exploratory data analysis, multiple regression models were trained and evaluated. Among the tested models, the Random Forest Regressor achieved the best predictive performance, demonstrating that nonlinear methods are effective for modeling complex fitness-related relationships.

Feature importance analysis showed that dream weight was the strongest predictor of actual body weight, while variables such as BMI, calories burned, heart rate, workout duration, and exercise intensity also contributed to model performance. Dream Weight is highly predictive of actual body weight and may act as a proxy for self-reported target weight, which could reduce the relative importance of physiological and exercise-based variables.

Overall, the results demonstrate that machine learning models can effectively use fitness and physiological data to predict body weight outcomes.

Acknowledgments

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References

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